Faculty of Computing and Information Technology-Rabigh,
(King Abdulaziz University)

Mobile-Based Application For Attendance Management System In The University Classes

Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Science in

(Computer Science)

BY

Abdulaziz Othman (1935680)

Ali Almejalli (2244549)

Alwaleed Alghamdi (2342305)

Supervised By: Dr. Nashwan Alromema

Fall2026

بسم الله الرحمن الرحيم

In the name of Allah, the most merciful, the most beneficent

CERTIFICATE

This is to certify that this project report entitled “Mobile attendance” by (**Abdulaziz Othman**), (**Ali Almejalli**), and (**Alwaleed Alghamdi)** submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science (CS) of King Abdulaziz University, Rabigh, during the academic year 2025-2026, is a Bonafede record of work carried out under our guidance and supervision. The results embodied in this report have not been submitted to any other University or Institution for the award of any degree or diploma.

Supervisor	**Head of the Department**
Dr. Nashwan Alromema	Dr. Hatem Almezarqi

DEDICATION

To our beloved parents

ACKNOWLEDGMENTS

We would like to express our special thanks of gratitude to our Head of Department *Dr. Hatem* as well as our supervisor *Dr. Nashwan* who gave us the golden opportunity to do this wonderful project on the topic “Smart support system for summer semester scheduling at FCITR “.

Secondly, we would also like to thank our parents and friends who helped us a lot in finalizing this project within the limited time frame.

ABSTRACT

Taking attendance of the students in the lectures is one of the tasks for the faculty members at the universities, and education organization. The time taken to do this task is between 10 to 15 minutes. This time period (10 to 15 minutes) also depends on the number of students present in the class, and sometimes depends on the area of the classroom, or theater room where the class is taken. The loss of this time will have an effect on the educational process, as well as the performance of the faculty members who are suffering from this lost time, especially in those lectures with more than 70 students. With the development of technology and the ease of using applications on smart devices, including watches, phones and tablets, it becomes necessary to benefit from this technology for solving this problem. In this project, we will develop a mobile-based application that works in the environment of smart devices to take the attendance of the present students in an intelligent and easy way so that the student can prepare himself to use the client-side App and by following the instructions of the faculty member.

Keywords: Attendance, Mobile, Application.

ABSTRACT (ARABIC)

ملخص

يعتبر تحضير الطلاب في المحاضرات إحدى المهام التي يقوم بها أعضاء هيئة التدريس في الجامعات والمؤسسات التعليمية، ويتراوح الوقت المستنفذ في تحضير الطلاب بين 10 إلى 15 دقيقة ويعتمد هذا الوقت (10الى 15 دقيقة) على عدد الطلاب في القاعة الدراسية او المدرج للأعداد الكبيرة من الطلاب. حيث يعاني بعض أعضاء هيئة التدريس من هذا الوقت المهدور، وتحديداً في المحاضرات التي يزيد عدد طلابها عن 70 طالباً، مع تطور التكنولوجيا وسهوله استخدام التطبيقات على الأجهزة الذكية بما فيها الساعات والهواتف والتابلت سوف نقوم بهذا المشروع بتطوير تطبيق يعمل في بيئة عمل الاجهزة الذكية لتحضير الطلاب بطريقه ذكيه وسهله بحيث يقوم الطالب بتحضير نفسه وذلك عن طريق متابعة التعليمات الصادرة من عضو هيئه التدريس.

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LIST OF ABBREVIATIONS

DBMS	-	Database Management System
ERD	-	Entity Relationship Diagram
GUI	-	Graphical User Interface
SQL	-	Structured Query Language
SDLC	-	software development life cycle
UML	-	Unified Modeling Language
RAM	-	random access memory

CHAPTER 1

INTRODUCTION

1.1 Introduction

Attendance tracking is an essential part of the educational process, ensuring that students participate regularly and maintain academic discipline. Traditionally, attendance is recorded manually by instructors—a process that can be time-consuming, error-prone, and inefficient, especially in large classrooms. With the increasing number of students in universities and educational institutions, this manual method often consumes a significant portion of lecture time that could otherwise be used for teaching.

In recent years, the rapid advancement of mobile technologies has created new opportunities to simplify and automate such routine tasks. A **Mobile Attendance System** aims to leverage the capabilities of smartphones and other smart devices to provide an intelligent, efficient, and user-friendly solution for managing student attendance. This system allows faculty members to record attendance digitally, while students can confirm their presence through a connected mobile application. The result is a faster, more accurate, and reliable attendance process that enhances both teaching efficiency and the overall classroom experience.

1.2 Problem Background

In educational institutions, recording student attendance is a routine yet essential task used to track participation and academic engagement. Traditionally, this process is performed manually—either by calling out names, passing around a sign sheet, or using Excel sheets. Although these methods are simple and familiar, they have several disadvantages. They consume a considerable amount of class time, especially in lectures with a large number of students, and are prone to human errors such as incorrect entries, proxy attendance, or misplaced records.

Faculty members often spend between 10 to 15 minutes of each session taking attendance using paper sheets or Excel files. This time could otherwise be devoted to teaching or discussion, and the repetitive nature of the task can reduce overall efficiency. Furthermore, maintaining attendance records in Excel files or on paper poses challenges in terms of organization, data management, and long-term storage. Retrieving or analyzing attendance data later can also be time-consuming.

With the rapid growth of mobile technology, smart devices such as smartphones and tablets have become powerful tools for automating everyday administrative tasks. By leveraging these technologies, the attendance process can be modernized and simplified. A **Mobile Attendance System** provides an effective solution to the issues of time consumption, inaccuracy, and inefficiency by offering a digital, automated, and user-friendly platform for both instructors and students.

1.3 Problem Statement

The existing methods for recording student attendance in universities, such as the use of paper-based sheets and Excel files, are inefficient, time-consuming, and susceptible to human error. Faculty members frequently spend a considerable portion of lecture time—often between ten to fifteen minutes—manually marking attendance, particularly in courses with a large number of students. This practice not only reduces the effective instructional time but also compromises the accuracy and reliability of attendance records. Furthermore, the manual management and storage of such records increase the likelihood of data misplacement and complicate the process of retrieval and analysis. Consequently, there is a pressing need for a **mobile-based attendance system** that automates the recording process, enhances accuracy, saves time, and ensures secure and efficient management of attendance data.

1.4 Research Question

There are some questions that we must ask before we start our project:

1) Who is the target group and the beneficiaries of this project?
2) Does this project matter to the academic community?
3) Does the project improve the educational process and faculty performance?

1.5 Research Aim

The primary aim of this research is to design and develop a **mobile-based management attendance system** that streamlines and automates the process of recording student attendance in academic environments. The system seeks to eliminate the inefficiencies of traditional paper-based and Excel-based methods by utilizing mobile technology to provide a faster, more accurate, and user-friendly solution. Specifically, the project aims to enable professors to efficiently manage attendance records, allow students to confirm their presence seamlessly, and maintain organized, secure data linked to courses and classrooms. Through this system, the study aims to enhance time efficiency, reduce human error, and improve the overall management of attendance in educational institutions.

1.6 Project Objectives

The objectives of the proposed project are as follows:

1) To develop a mobile-based system for automating student attendance.

2) To reduce the time and effort spent on manual attendance.
3) To improve the accuracy and reliability of attendance records.
4) To provide a simple and user-friendly interface for professors and students.
5) To store and manage attendance data securely and efficiently.

1.7 Research Scope

The assumptions of the work in this project are limited to the following:

1. What the system will do (In Scope):
    - Allow professors to take attendance digitally using a mobile app.
    - Allow students to confirm their presence through the app.
    - Record attendance data with date and time.
    - Store attendance information in a database.
    - Allow professors to view or download attendance reports.
2. What the system will not do (Out of Scope):
    - It will not create or manage courses
    - It will not handle grading, exams or performance tracking

1.8 Significance of the Study

The **Mobile Attendance management System** transforms educational efficiency by replacing time-consuming manual attendance with an automated mobile solution. This directly addresses the loss of 10-15 minutes of instructional time in each lecture, particularly benefiting large classes.

For our 3-member team, this project represents a significant capstone achievement, allowing us to practically apply our combined skills in mobile development, database design, and system architecture to create a real-world solution.
The system delivers key benefits:

- **Faculty** save valuable teaching time and reduce administrative work
- **Students** enjoy a seamless attendance process using familiar devices
- **Institutions** obtain a scalable, modern solution without major investment

This project demonstrates how strategic technology applications can enhance educational processes while providing essential hands-on experience for our transition into professional software development careers.

1.9 Conclusion

This chapter has highlighted the challenges faced by educational institutions in managing student attendance using traditional methods such as paper sheets and Excel files. These approaches are time-consuming, prone to errors, and inefficient, especially in large classrooms. The manual process not only reduces valuable teaching time but also complicates data storage and retrieval.

With the widespread use of mobile technology, there is a strong opportunity to improve this process through automation. The proposed Mobile Attendance System offers a practical solution by enabling faculty to record attendance digitally and allowing students to confirm their presence using smart devices. This system aims to save time, improve accuracy, and streamline attendance management.

For our development team, this project represents a meaningful application of our technical skills and a chance to contribute to the modernization of academic processes. By addressing a real-world problem with a user-friendly and efficient solution, the project supports better classroom management and enhances the overall educational experience.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Attendance tracking is an important part of education, helping to ensure that students participate regularly and stay engaged in their studies. Traditionally, teachers record attendance by hand, which can be slow, prone to mistakes, and inefficient, especially in large classes. As the number of students in universities and schools increases, this manual method often takes up valuable time that could be spent on teaching.
Recent advancements in mobile technology offer a new way to simplify and automate attendance tracking.

A Mobile Attendance System uses smartphones and other smart devices to create an easier and more efficient way to manage student attendance.

This system allows teachers to record attendance electronically, while students can confirm their presence through a connected mobile app. The result is a faster, more accurate, and reliable attendance process that enhances teaching and improves the classroom experience.

This literature review will look at how attendance tracking methods have evolved, the impact of mobile technology on education, and the benefits and challenges of using Mobile Attendance Systems. By exploring current research and examples, we aim to understand how these innovative solutions can improve teaching efficiency and enrich the overall educational experience.

2.2 Background of Related projects/research

This section explores notable projects and research initiatives that have significantly influenced the development of automated attendance systems. These implementations have demonstrated practical effectiveness and widespread adoption across various sectors.

1. Biometric attendance systems: the use of unique human biometric attributes, as fingerprints or facial and voice recognition to automate attendance tracking which used in many different companies and institutions.
2. QR code-based systems: using QR codes for attendance tracking which works by scanning distinguished QR code to recognize the user, also used in many universities and many different institutions.

2.3 Information about similar projects/research

Mobile-based phone attendance system, the system is called MPBAS (Mobile Phone Based Attendance System)
It is designed to help lecturers take attendance easily, securely, and with fewer errors, The application is built using Android technology and connects to a central server, Lecturers log into the app, access the attendance list, and record student presence digitally.

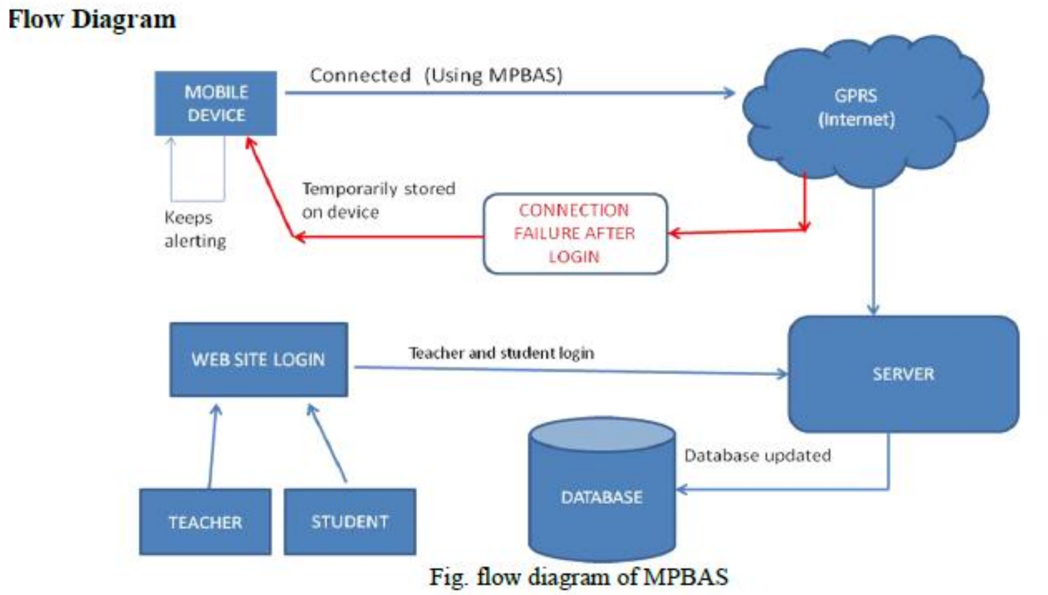

Figure 2.1 Flow diagram of MPBAS

2.3.1 Project/ research (1)

Mobile Attendance System Using QR Codes Technology, the research aims to improve traditional attendance methods by introducing a mobile-based system that uses QR codes. This approach is designed to be simple, fast, and cost-effective, especially for large classrooms.

2.4 Comparison of the projects

Table 2.1: The comparisons of our proposed project with existing ones

Main functionalities	**Projects**		
	Online Attendance Management System	Attendance Management System	Mobile-Based Application For Attendance Management System In The University Classes
The interface is easy to use	√	√	√
Provide Scheduling process	√	√	√
Provides Self-service for students	√		√
Student-Self registration	-	-	√
Provides Mobile application	-	-	√
Paying Fees	-	-	-

2.5 Conclusion

This chapter has reviewed the evolution of attendance tracking methods, ranging from traditional manual approaches to modern automated systems. The discussion highlighted the limitations of manual recording, such as inefficiency and susceptibility to errors, and examined how technological innovations — including biometric systems, QR code solutions, and mobile-based applications — have sought to address these challenges.

Existing projects, such as biometric and QR code-based attendance systems, demonstrate the potential of automation in improving accuracy and reliability. Similarly, mobile-based systems like MPBAS illustrate how smartphones and connectivity can simplify attendance management for lecturers and students. However, the comparative analysis revealed that while these systems provide core functionalities such as ease of use and scheduling, they often lack extended features like student self-service, self-registration, and integration with broader academic processes.

The findings underscore the need for a more comprehensive mobile attendance solution that not only records student presence efficiently but also enhances the overall educational experience by reducing administrative workload and fostering greater student engagement. By addressing the gaps identified in existing systems, the proposed project aims to contribute to the development of a robust, scalable, and user-friendly mobile attendance system tailored to the needs of modern educational institutions.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter describes the research methodology adopted for the development of the Mobile Attendance System. The methodology provides a structured approach to ensure the system is designed, implemented, and evaluated effectively to address the challenges of traditional attendance tracking methods.

The project follows the **System Development Life Cycle (SDLC)**, a widely used framework for developing software solutions. Each phase of the SDLC—planning, analysis, design, implementation, testing, and deployment—is carefully outlined to ensure the project meets its objectives with precision and efficiency.

3.2 Methodology

The methodology of records system improvement is the systematic description, explanation, and assessment of all factors of methodical data system development. The methodology is the aggregate of paradigm, method, procedure, rules, technique, and tools with programming languages that will be used for evaluation and to layout the structure of a system. Some examples of methodology fashions are the waterfall model, spiral model, and prototype model. Figure 3.1 suggests the methodology and the phases for our challenge development.

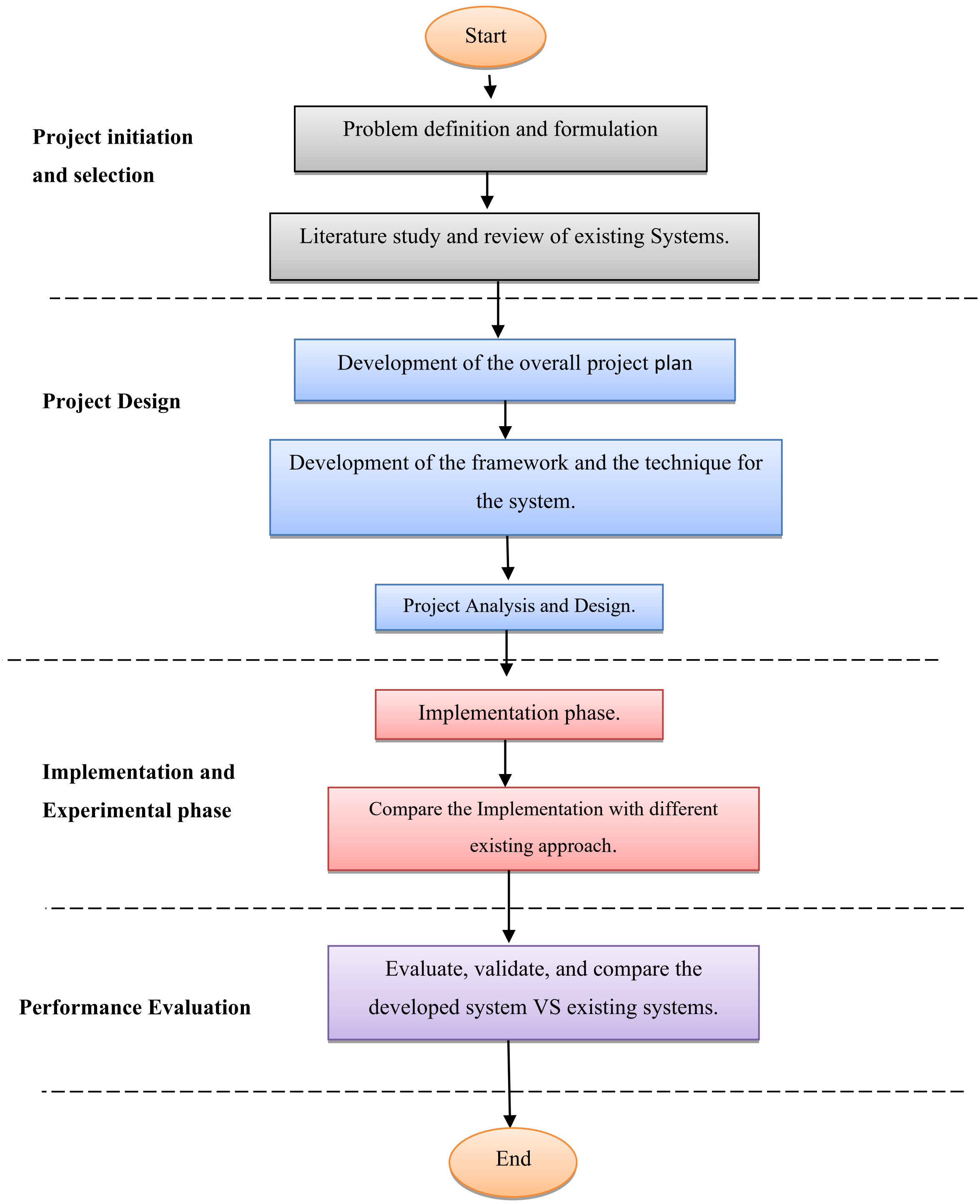

Figure 3.1 The general framework for the project development

3.3 Justification of Methodology

The System Development Life Cycle (SDLC) was chosen as the methodology for this project because it provides a structured and systematic approach to software development. The SDLC ensures that the project progresses through clearly defined phases—planning, analysis, design, implementation, testing, and deployment—each contributing to the successful delivery of a reliable and user-friendly system.

3.3.1 Methodology Prototype

In this system, the Evolutionary Prototype used to be chosen to endure the procedure of creating the system. Each phase wanted to be accompanied to boost a well-planned system. The six phases in this methodology are as below:

1) Project Planning Phase.
2) Analysis of Project Requirements Phase.
3) Prototype Design Phase.
4) Coding and Prototype Development Phase.
5) Testing Phase.
6) Updating and Maintenance Phase.

3.3.2 Project Planning Phase

The project planning phase is to clarify the background problems and activity related based on the proposed system. Background problems, objectives, and scope of the project are to be defined and clarify to have a better understanding.

3.3.3 Analysis of Project Requirements Phase

An evaluation of the project requirements phase, search about the contemporary machine to discover problems that appear to gain the goal of the project. Findings and research for the technology used and methodology that can be used to resolve the trouble for device development should be applied as well. The research and findings are based on books and records from websites. The data accumulated will prepare in an ideal manner and will be analysed or as references with the aid of the developer.

3.3.4 Project design Phase

The prototype format section is based on the structural thinking of the float sketch of the system, so it will be less complicated to be understood in the coding process. This phase will put into effect the interface plan of a machine regardless of functions.

3.3.5 Coding and Prototype Development Phase

In this phase, it is a procedure of implementing the coding according to the functions of the machine into the last version of the prototype that is developed. Each of the machine modules will be developed according to specification and architecture design.

3.3.6 Testing Phase

The testing phase is a system to look at and check the functional system that has been developed whether the device fulfils the requirement of clients. This process is also to identify and look at whether any bugs located in the system. There will be corrections made to the functions and coding if any or add new functions if needed. This method will loop until the prototype developed to turn out to be totally functional system according to specification, objective, and scope of the project.

3.8 Updating and Maintenance Phase

Updating and maintenance phases are the strategies of updating the device when it is out of specification noted or to maintain the wholly functional device the place necessary. Updating and renovation is the remaining segment to enhance the device is used if there are new necessities wanted to be brought in.

3.9 Justification of Choosing Evolutionary Prototype

The purpose evolutionary prototype is chosen based on advantages as below:

1) This methodology allows repeat of the improvement phases to pick out the requirements which are no longer stated.
2) This methodology is bendy as it can enhance the communication between developer and client. Changes that may be made from time to time are no longer in trouble to fulfil the requirements of the client.
3) The adjustments in the prototype are to be greater environment friendly and less difficult to manage.
4) This methodology let developer put in greater effort and dedication to produce a wholly functional system.
5) The nice of requirements and specifications can improve because the early determination of what the consumer needs can result in quicker and much less pricey software.

3.10 Analysis of System Requirements

Analysis and specification necessities of a system are necessary to become aware of the hardware and software program necessities in the method of creating a machine to allow the method to be applied in a right and efficient manner. Software and hardware chose to have to be suitable for the method of system improvement so keep away from issues such as value and misuse of the hardware's functionality.

3.11 Justification of Hardware

The justification of hardware necessities is necessary to develop a Student's course registration applications. Below are the smartphone necessities for the client, and the Computer necessities for the developer. Table 3.1 indicates the type, minimal requirement, and advocated requirement.

Table 3.1: Hardware Requirements for Client's Phone

Client Requirement		
Type	**Minimum**	**Recommended**
Processor	Version 8.0 (Oreo)/ : Version 12.0	Version 10.0 (Q)/ Version 14.0
Memory (RAM)	2GB	4GB
Screen Resolution	5 in	6 in or higher
Memory Space	2 GB	8 GB or higher

Table 3.2: Hardware Requirements for Developers Computer

Developer Requirement	
Type	**Recommended**
Processor	Intel Core i7-12700KF Desktop Processor 12
Memory (RAM)	16 GB
Screen Resolution	1080x1920
Hard Disk Space	250GB

3.12 Justification of Software

The justification of a software program is primarily based on software wished to enforce in the technique of device development. Suitable software will ease builders to increase a system in a superb and environment-friendly way.

Table 3.3 Software Developer Requirements

Software Developer Requirement	
Operating System	Windows 10
Integrated Development Environment	Oracle Application Express (APEX)
Database type	Oracle database + PL/SQL+ APEX

3.13 Microsoft Windows 10

Windows 10 has high compatibility with most cutting-edge software tools and applied sciences nowadays. It is a layout for users to maximize the working gadget's overall performance and functions as well.

3.14 Oracle Application Express (APEX)

Oracle Application Express is a web-based software development environment that runs on an Oracle database. It is fully supported and comes standard with all Oracle Database editions and, starting with Oracle 11g, is installed by default as part of the core database install.

3.15 Database type

Oracle Database is a multi-model database management system produced and marketed by Oracle Corporation. It is a database commonly used for running online transaction processing, data warehousing, and mixed database workloads.

3.16 Draw.io

Draw.io is an internet site used to draw diagrams, and it is free to use with multiple preferences in designing diagrams because it is a website we saved the time wanted to set up a sketch format software which is additionally now not free to use.

3.17 Android Studio

Android Studio is the official integrated development environment **(IDE)** for building Android apps, It gives you code editor, emulator, debugger, and UI designer in one place, and it’s backed by google to stay up to date.

3.18 Microsoft Word

Microsoft Office Word is used to document the report.

3.19 Conclusion

The Evolutionary Prototype methodology was once chosen because it enables a repeat of the development phases to discover the requirements which are now not stated. In conclusion, the Evolutionary Prototype method enables flexibility in the development process which can amplify the accuracy in fulfilling client requirements.

CHAPTER 4

System Analysis and Design

4.1 Introduction

The System Analysis and Design phase represents a critical milestone in the development of the Mobile Attendance System, where theoretical concepts are transformed into practical, implementable solutions. This chapter provides a comprehensive examination of the systematic approach employed to analyze the current attendance management challenges and design an efficient mobile-based solution that addresses the identified requirements

4.2 Software Requirements

This section outlines the essential software requirements for the development and deployment of the Mobile Attendance System. The system is designed to streamline and automate the process of recording student attendance using mobile technologies, thereby enhancing efficiency, accuracy, and user experience in educational environments.

4.2.1 Functional Requirements

The functional requirements are an important step in system analysis and design:

- System must allow secure login for faculty and students using unique credentials (Name , ID)
- Students should be able to mark attendance using a session code or QR-code.

- The system should notify users of session start times, missed attendance, or updates.
- The system must be able to record only the student who in the class.
- Instructors must be able to manage, create and delete sessions such as (subjects, timings, activation codes/QR-codes)

4.2.2 Non-Functional Requirements

- System must handle large numbers of students and sessions simultaneously without delay.
- System should have intuitive and mobile friendly interface.
- System must have clear error handling messages to guide the user
- System must have at least two languages (Arabic, English)
- System should have a push notification for session start time, activation code start and end time.

The Conceptual Design (ER-Diagram)

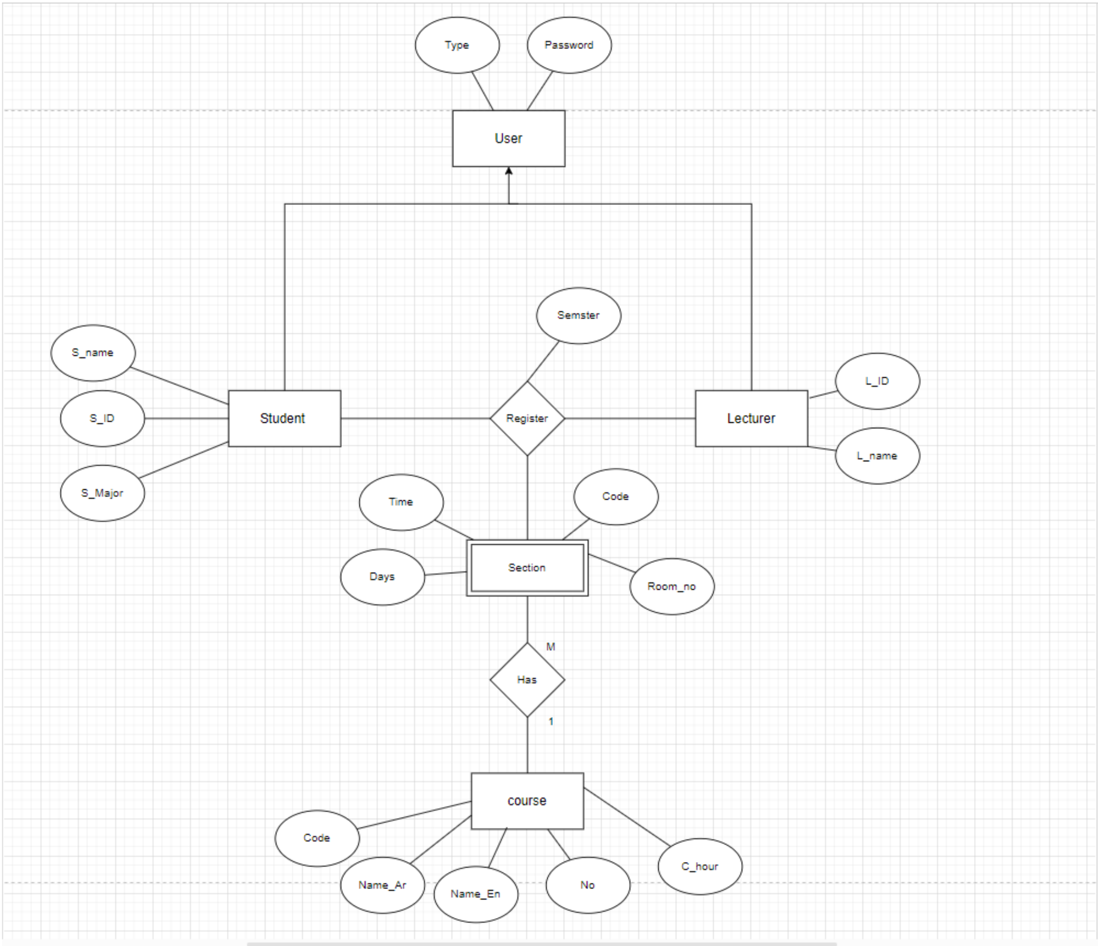

Figure 4.1: The Conceptual Schema (ERD) for the System

4.4 The Logical Design (Mapping)

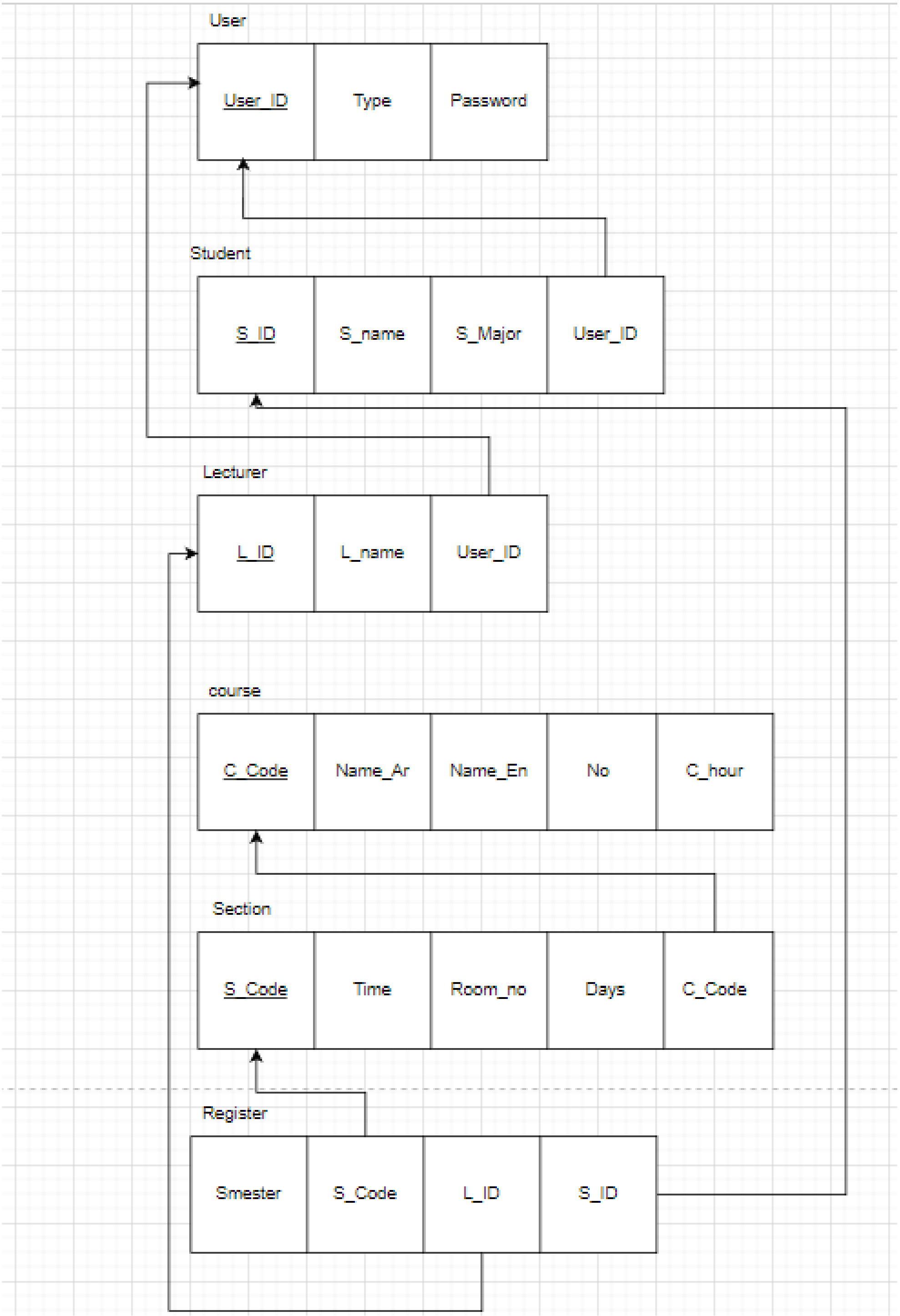

Figure 4.2: The Logical Schema (Mapping)

.5 Use case Diagram

A use case diagram is a graphic depiction of the interactions among the elements of the system; Figure 4.3 depicts the different actors and functionality of the system.

Figure 4.3: The Use case Diagram

4.6 Activity Diagram

The activity diagram is another important diagram in UML to describe the dynamic aspects of the system, Figure 4.4 illustrates these aspects.

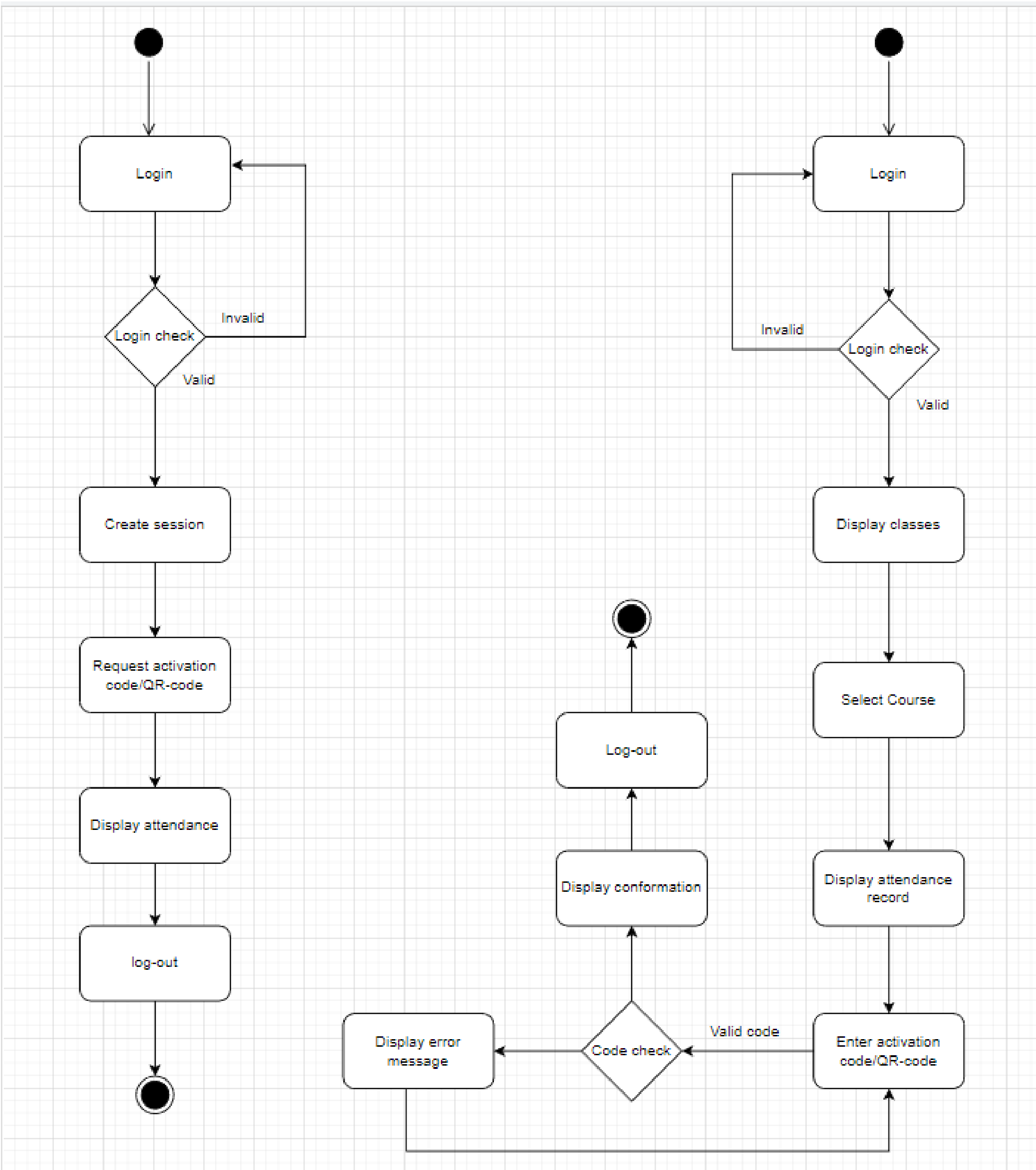

Figure 4.4: The Activity Diagram

4.7 Sequence Diagram

A sequence diagram as in Figure 4.5 shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario. Sequence diagrams are sometimes called event diagrams or event scenarios.

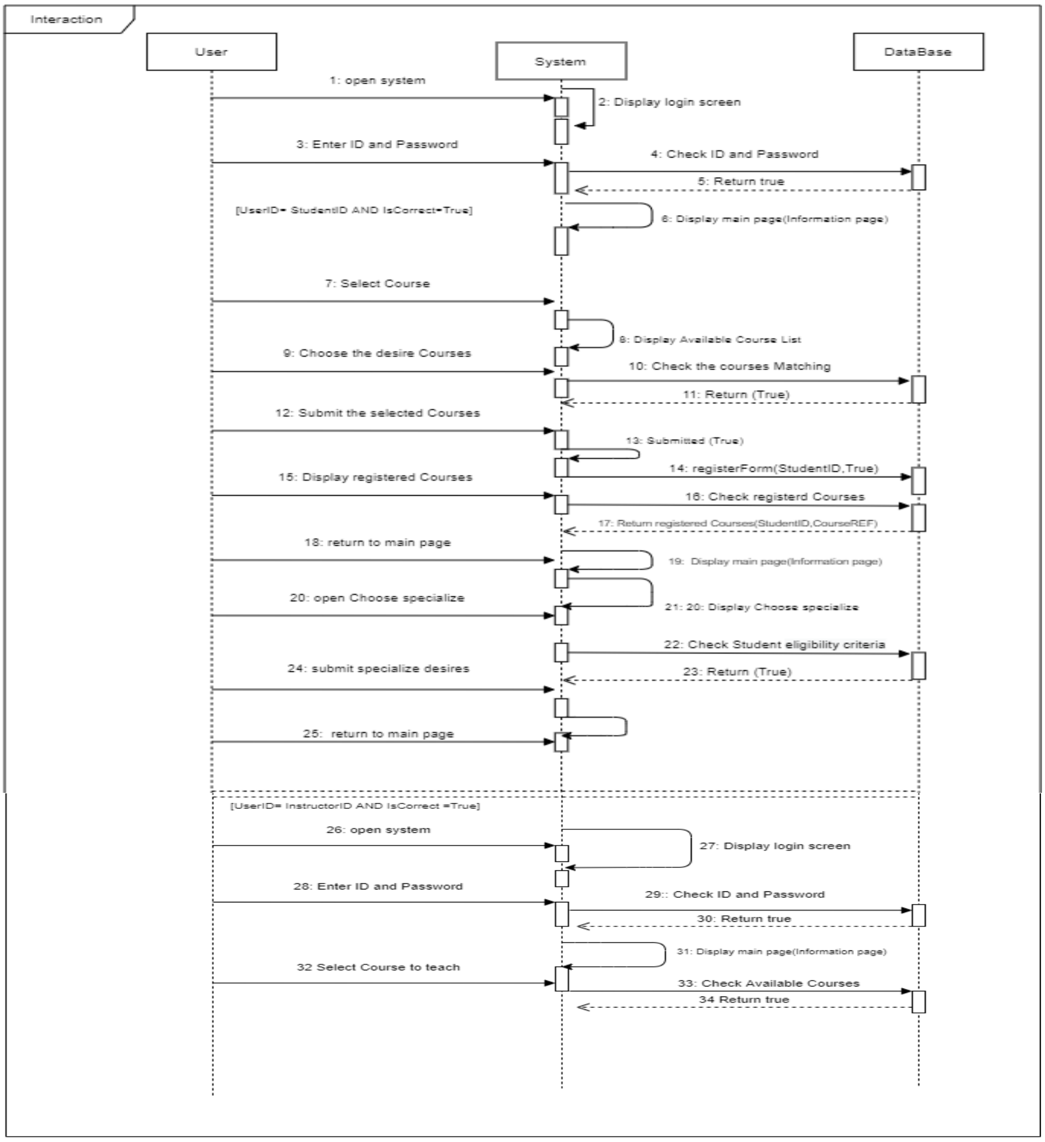

Figure 4.5: The Sequence Diagram

4.8 Architecture Design of the System (Prototype)

Once the requirements analysis phase was completed, we transitioned into the design phase to gain a sharper and more structured vision of the project before moving into implementation. The following subsections highlight the different layouts of the system's components, offering a clearer roadmap for development.

4.8.1 Log-In Screen

The login page contains the ID of student/Instructor and the password, each of them can check if a student or Instructor. The system also is bilingual.

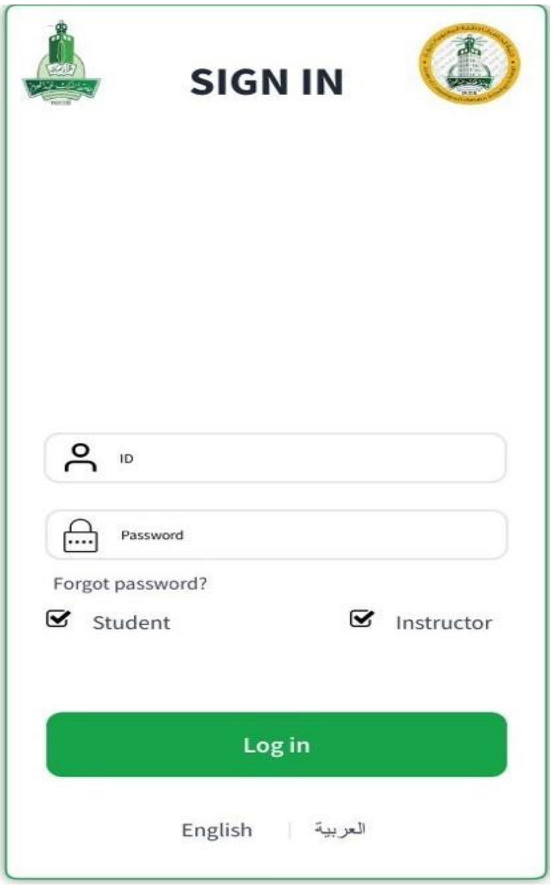

Figure 4.6: Log-In screen

4.8.2 Student Login page

This page allows Students to check in the checkbox to determine whether it is a student or an Instructors.

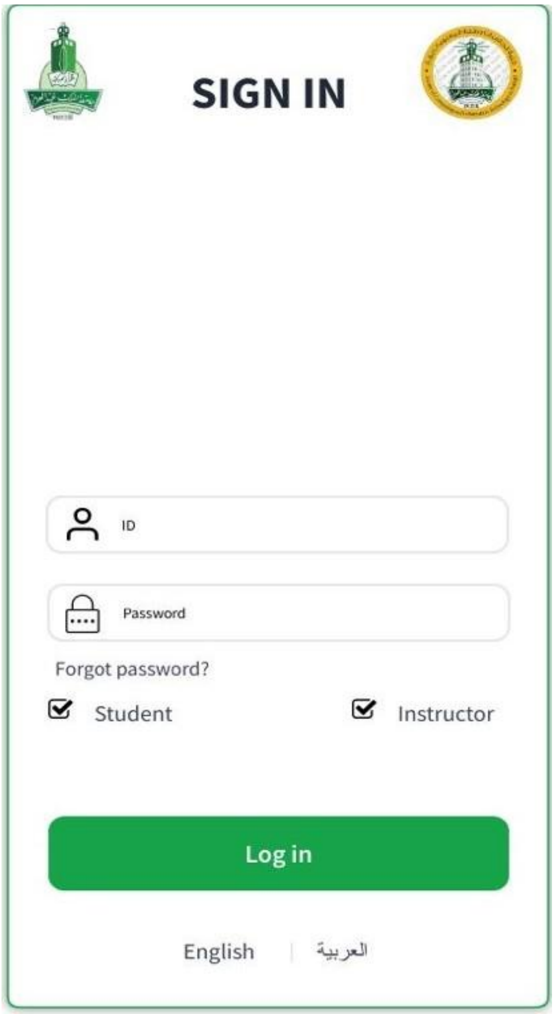

Figure 4.7: Student login page

4.8.3 Student Profile Page

This page contains the student information and the services provided by our system as select courses also show the registered courses after the student chooses it, moreover, allows the new student who not specialized to choose the specializations desires. If the student is specialized and wants to change the specializations, he can change it easily.

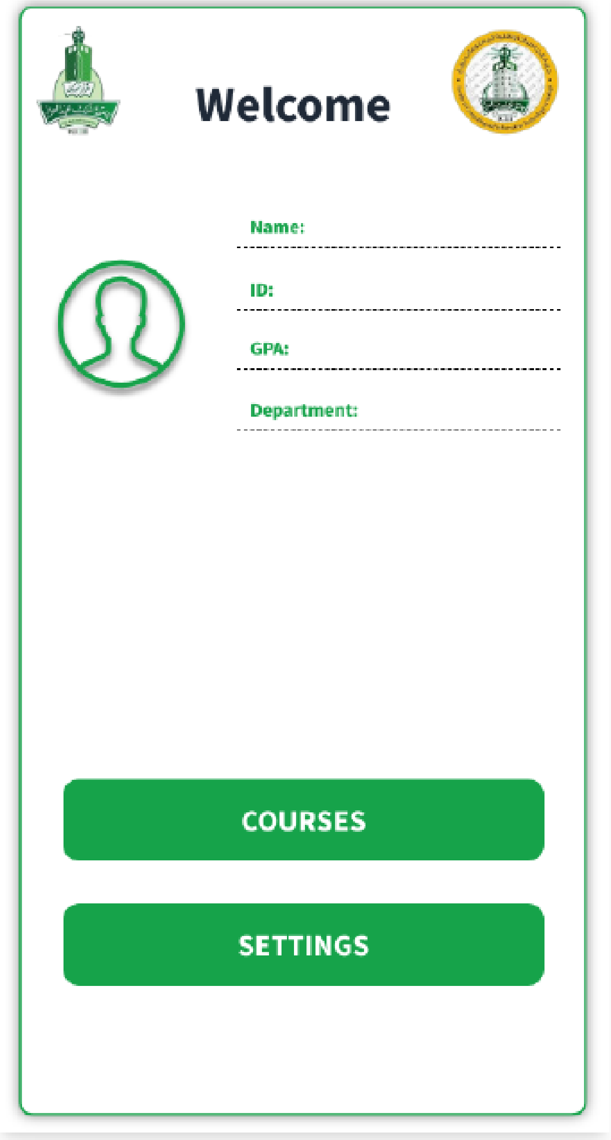

Figure 4.8: Student Profile Page

4.9 Conclusion

This chapter has successfully translated the project's objectives into a detailed technical blueprint for the Mobile Attendance System. We defined the system's core functions and quality standards through software requirements, established its data structure with ER and logical diagrams, and modelled its dynamic behaviour using UML (Use Case, Activity, and Sequence diagrams). The provided prototype designs offer a clear vision for the user interface. Collectively, this analysis and design provide a solid and direct foundation for the system implementation phase.

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 Introduction

System implementation is a critical phase in the software development lifecycle where the design specifications are transformed into a working, functional system. This chapter details the implementation of the Attendance Management System, covering the practical aspects of how the system was developed, built, and deployed. The importance of this phase lies in bridging the gap between theoretical design and practical execution, ensuring that all system components work together seamlessly to achieve the project objectives. Throughout this chapter, we will discuss the development environment, the tools and technologies selected, the system architecture in practice, and the integration of all components to create a fully functional attendance management system. Proper implementation ensures that the system is robust, maintainable, and meets the performance requirements necessary for deployment in a university environment.

5.2 Implementation

The implementation of the Attendance Management System was carried out using an evolutionary prototype methodology, which allowed for iterative development and continuous integration of feedback throughout the development cycle. This approach enabled the team to build the system incrementally, refining the design based on practical development experiences. The implementation process involved three main layers that work together cohesively: the mobile client layer, the API layer, and the database layer.

The mobile client was developed using Kotlin, a modern programming language that provides null-safety and reduces boilerplate code, making it ideal for Android development. The API layer was built using Oracle Application Express (ORDS), which provides a RESTful interface to database operations through PL/SQL procedures. The database layer utilizes Oracle Database to store and manage all system data including users, courses, sections, enrollments, attendance codes, and attendance records.

During the implementation phase, each component was developed independently before being integrated with the other components. The mobile application uses Retrofit 2.11.0 for HTTP networking and Gson 2.10.1 for JSON serialization, enabling seamless communication with the API layer. The API layer exposes multiple endpoints that handle authentication, course management, attendance session management, and attendance recording. This layered approach to implementation allowed for parallel development of components while maintaining clear interfaces between them, resulting in a system that is modular, maintainable, and scalable.

5.3 Development Environment

The development environment encompasses all the hardware, software, and infrastructure resources necessary to develop, build, and deploy the Attendance Management System. A well-configured development environment is essential to ensure smooth development workflows, effective collaboration among team members, and successful system deployment. The AMS development environment consists of two main components: the server infrastructure that hosts the backend API and database, and the software tools and frameworks used for development. These components work together to provide a complete ecosystem for building, testing, and running the attendance management system..

5.3.1 Server

The server infrastructure for the mobile-based management attendance system is hosted on Oracle Cloud Infrastructure (OCI) in the jeddah region, which is geographically located in Jeddah, Saudi Arabia. This ensures low latency and compliance with local data residency requirements. The server runs Oracle Application Express (ORDS - Oracle REST Data Services), which acts as the bridge between the mobile application and the database.

ORDS exposes PL/SQL procedures and SQL queries as RESTful web services, allowing the mobile client to communicate with the database through HTTP/HTTPS requests.

All API endpoints use HTTPS protocol for secure communication, and every response is wrapped in a JSON envelope format to ensure consistent data handling across the system. The base URL for all API endpoints is: https://tfjudkfbikoeqek-studentaiprojects.adb.me-jeddah-1.oraclecloudapps.com/ords/gp2user2/attendance_api/.

The server infrastructure provides reliable, scalable hosting that can handle multiple concurrent requests from students and instructors accessing the system simultaneously.

5.3.2 Software

The software stack used for developing the mobile-based management attendance system includes several key technologies selected for their reliability, performance, and suitability for the project requirements. Android Studio serves as the integrated development environment for mobile application development, providing tools for designing user interfaces, debugging, and testing. Kotlin was chosen as the primary programming language for the Android client due to its modern features including null-safety, extension functions, and reduced boilerplate code compared to Java.

On the backend, Oracle Database 12c+ provides the persistent data storage layer, ensuring data integrity and reliable query execution. Oracle Application Express (APEX) is used to develop the REST API endpoints that expose database operations as web services. For communication between the mobile client and the API, the application uses Retrofit 2.11.0 for HTTP networking and Gson 2.10.1 for JSON serialization and deserialization. The user interface follows Material Design guidelines using Material Components 1.12.0, providing a modern and intuitive user experience. Android's RecyclerView is utilized for efficient list rendering of courses and attendance records. These software components were selected to ensure compatibility, performance, and ease of maintenance throughout the system's lifecycle.

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5.4 Overall System Development

The overall system development of the mobile-based attendance system involves three main components: the Android mobile application, the Oracle Application Express (ORDS) REST API, and the Oracle Database. These components communicate through HTTPS requests and well-defined interfaces to create a functional system.

The mobile application sends user input to the REST API, which processes requests by executing PL/SQL procedures and SQL queries against the database. The database stores all essential information including user credentials, courses, enrollments, attendance codes, and attendance records. Changes made through the mobile application are immediately reflected in the database and available to authorized users.

The modular architecture allows each component to be developed independently while maintaining consistent interfaces, ensuring the system is scalable, maintainable, and flexible for future enhancements.

5.4.1 User Interface Development

The user interface of the mobile-based attendance system was developed using Android's View-based components to ensure compatibility and optimal performance on various devices. The application follows Material Design guidelines, utilizing Material Components 1.12.0 to provide a modern and intuitive user experience. The interface design incorporates the university's brand color (`#2E4F3D`) throughout the application for visual consistency.

The system has distinct user interfaces for students and instructors, each tailored to their specific needs. The student interface includes activities for login, home page displaying enrolled courses, course details, attendance marking, and attendance history. The instructor interface includes login, home page displaying taught sections, section details, attendance code generation with timer, and attendance history viewing. Navigation between activities is handled through Intent objects, allowing seamless transitions between different screens. RecyclerView components are used for efficient rendering of course lists and attendance records. The interaction between pages follows a logical flow that matches the user's workflow, from authentication through attendance management and history viewing.

5.4.2 Database Connection

The database connection is established through the Retrofit HTTP client, which communicates with the Oracle Application Express (ORDS) REST API endpoints. Retrofit 2.11.0 handles all HTTP requests and responses, while Gson 2.10.1 manages the serialization and deserialization of JSON data between the mobile application and the API layer. The connection uses HTTPS protocol to ensure secure transmission of sensitive data.

The database schema includes six main tables: USERS (storing user credentials and profiles), COURSES (course information), SECTIONS (section and instructor mappings), SECTION_ENROLLMENTS (student enrollments), ATTENDANCE_CODES (active session codes), and ATTENDANCE_RECORDS (attendance marks). All database operations are performed through PL/SQL stored procedures exposed via ORDS endpoints. Timestamps are stored in Asia/Riyadh timezone using CAST(SYSTIMESTAMP AT TIME ZONE 'Asia/Riyadh' AS TIMESTAMP) to ensure consistent time handling across the system without daylight saving complications.

5.4.3 Coding

The coding implementation uses Kotlin as the primary programming language for Android development, selected for its null-safety features and modern syntax. PL/SQL is used for backend stored procedures and database operations. The application implements a callback-based architecture where Retrofit's Callback interface handles asynchronous HTTP responses from the API. Each API response is wrapped in a JSON envelope containing a "body" property and "status_code," which the application parses to extract the actual data.

The attendance submission is implemented as an atomic operation through a single POST /attend/ endpoint that validates the code and records attendance in one transaction, preventing race conditions. The timing logic uses an 80/100 split where submissions before 80% of the session duration are marked as "Present" (P), submissions between 80% and 100% are marked as "Late" (L), and submissions after 100% are rejected as expired. The application uses SharedPreferences for local storage of user session data and application settings including language preference and notification settings.

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5.4.4 Authorization Scheme

The authorization scheme of the mobile-based attendance system is implemented entirely on the server side, ensuring secure access control and preventing unauthorized data manipulation. User authentication is performed through a login endpoint that validates email or ID against stored credentials in the USERS table. Upon successful authentication, the user's ID and role (student or instructor) are stored in SharedPreferences for session management.

The authorization is role-based, where students and instructors have different access permissions within the system. Students can view their enrolled courses, mark attendance for active sessions, and view their attendance history filtered by section. Instructors can view their taught sections, generate attendance codes with customizable durations, view attendance records for their sessions, and manage attendance sessions. All API endpoints validate the user's identity through HTTP headers and enforce appropriate authorization checks before executing database operations. This server-side authorization approach ensures that access control cannot be bypassed through client-side manipulation.

5.5 Conclusion

The implementation phase of the mobile-based attendance system successfully transformed the design specifications into a fully functional and integrated system. Through the use of modern technologies including Kotlin for mobile development, Oracle APEX for API development, and Oracle Database for data storage, a robust three-tier architecture was created that effectively separates concerns and allows for independent component development and maintenance.

The development environment provided a solid foundation with Oracle Cloud Infrastructure hosting the backend services and a comprehensive software stack including Retrofit, Gson, and Material Components. The user interface was carefully designed to provide an intuitive experience for both students and instructors, while the database connection was implemented securely through HTTPS communication with proper data serialization.

The coding implementation successfully addressed key challenges such as atomic attendance submission to prevent race conditions, timezone consistency across the system, and server-side authorization to ensure security. The modular architecture and careful implementation of each component resulted in a system that is scalable, maintainable, and ready for deployment in a university environment. The implementation demonstrates proper software engineering practices and provides a solid foundation for the system to meet its intended objectives of streamlining the attendance management process.

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CHAPTER 6

System Testing

6.1 Introduction

System testing is a critical phase in the software development lifecycle where the entire integrated system is tested as a whole to ensure it meets the specified requirements and functions correctly in its intended environment. This chapter documents the testing process undertaken for the mobile-based attendance system, covering the various testing activities performed to identify and resolve defects before deployment. The importance of system testing lies in ensuring that all components of the system work together seamlessly, that data flows correctly through all layers, and that the system behaves as expected under various conditions and scenarios. Through rigorous testing, potential issues were identified and corrected, resulting in a stable and reliable system that is ready for deployment in a university environment.

6.2 Testing Process

The testing process for the mobile-based attendance system was carried out throughout the development lifecycle, following the evolutionary prototype methodology adopted during implementation. Rather than treating testing as a separate standalone phase, testing was conducted continuously as each component and feature was developed, allowing issues to be identified and resolved early in the development process. This approach ensured that problems were caught at the component level before integration, reducing the complexity of debugging and fixing issues in the fully integrated system.

The testing process covered all major aspects of the system including the mobile application functionality, the REST API endpoints, the database operations, and the integration between all three layers. Each feature was tested against its requirements to verify that it behaved as expected under normal conditions as well as edge cases. Issues discovered during testing were documented, analyzed, and resolved before moving on to the next development iteration. This iterative approach to testing resulted in a progressively more stable and reliable system throughout the development cycle.

6.2.1 Unit Testing

Unit testing was performed on individual components of the mobile-based attendance system to verify that each component functions correctly in isolation. On the mobile client side, individual activities and their interactions with the API were tested to ensure correct behavior. On the backend side, each PL/SQL procedure and REST API endpoint was tested independently to verify that it produces the correct output for a given input. This level of testing allowed the development team to identify and resolve issues at the component level before integration, ensuring that each unit of the system behaves as expected before being combined with other components.

6.2.2 Integration Testing

Integration testing was conducted to verify that the different components of the mobile-based attendance system work together correctly. This involved testing the communication between the Android mobile application and the Oracle APEX REST API, as well as the interaction between the API layer and the Oracle Database. Key integration points that were tested include the authentication flow between the mobile client and the server, the attendance code generation and validation process, and the correct flow of data from the mobile application through the API and into the database. Integration testing ensured that data is correctly serialized and deserialized between layers and that all components interact as expected.

6.2.3 System Testing

System testing was performed on the fully integrated mobile-based attendance system to verify that it meets all specified requirements and functions correctly as a complete system.

This involved testing the entire attendance workflow from end to end, including the instructor generating an attendance code, students submitting the code, the system correctly marking attendance as Present or Late based on the 80/100 timing split, and the automatic marking of absent students when the session closes.

System testing also verified that the role-based access control functions correctly, ensuring that students and instructors can only access features appropriate to their role. Additional system-level tests verified that notifications are delivered correctly, attendance history is displayed accurately, and session cancellation works as expected.

6.2.4 User Acceptance Testing

User acceptance testing was conducted to verify that the mobile-based attendance system meets the needs and expectations of its intended users, both students and instructors.

Real users interacted with the system to perform typical tasks such as logging in, viewing courses, generating attendance codes, marking attendance, and viewing attendance history.

Feedback gathered during this phase was used to refine the user interface and improve the overall user experience. The testing confirmed that the system is intuitive and easy to use, that the attendance workflow is clear and straightforward, and that the system effectively addresses the problem of time-consuming manual attendance taking in university classrooms.

6.3 Testing

6.3.1 Login Page

The login page allows users to authenticate using either their university email or student ID along with their password. The system automatically identifies the user's role upon successful login and redirects them to the appropriate home page, whether student or instructor.

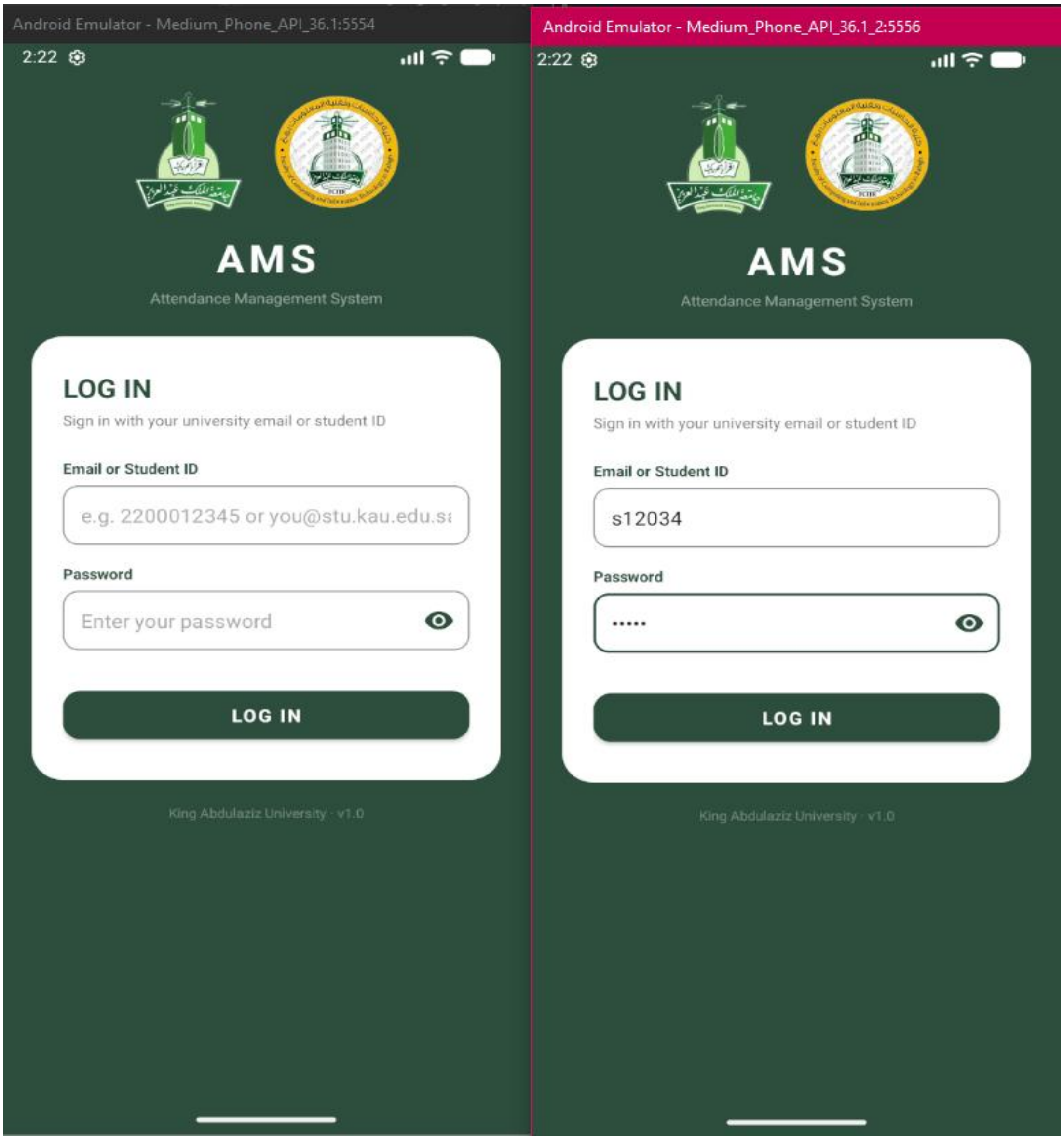

Figure 6.1 login Page

6.3.2 Home Page

Upon successful login, users are directed to their respective home pages based on their role. The instructor home page displays their ID, department, and list of assigned classes, along with options to take attendance or view history. The student home page displays their GPA, ID, major, and enrolled courses, with options to mark attendance or view their attendance history..

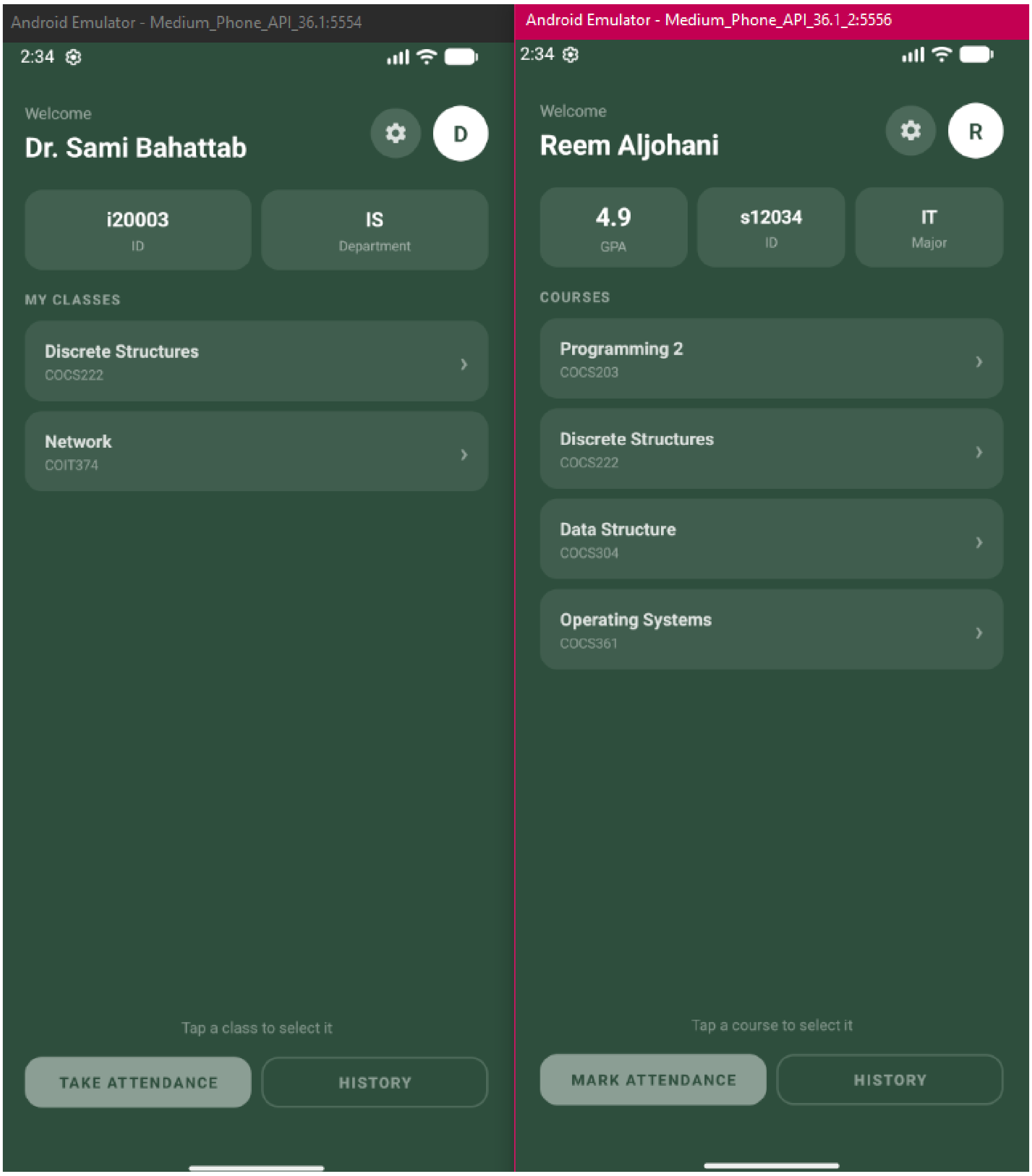

Figure 6.2 Home Page

6.3.3 Attendance Session

The attendance session screen showcases the core functionality of the system from both perspectives simultaneously.

The instructor's screen displays the generated attendance code along with a session timer counting down, and a slide to cancel option to end the session early.

The student's screen displays the same code which was automatically filled via notification, and a submit button to mark their attendance for the active session.

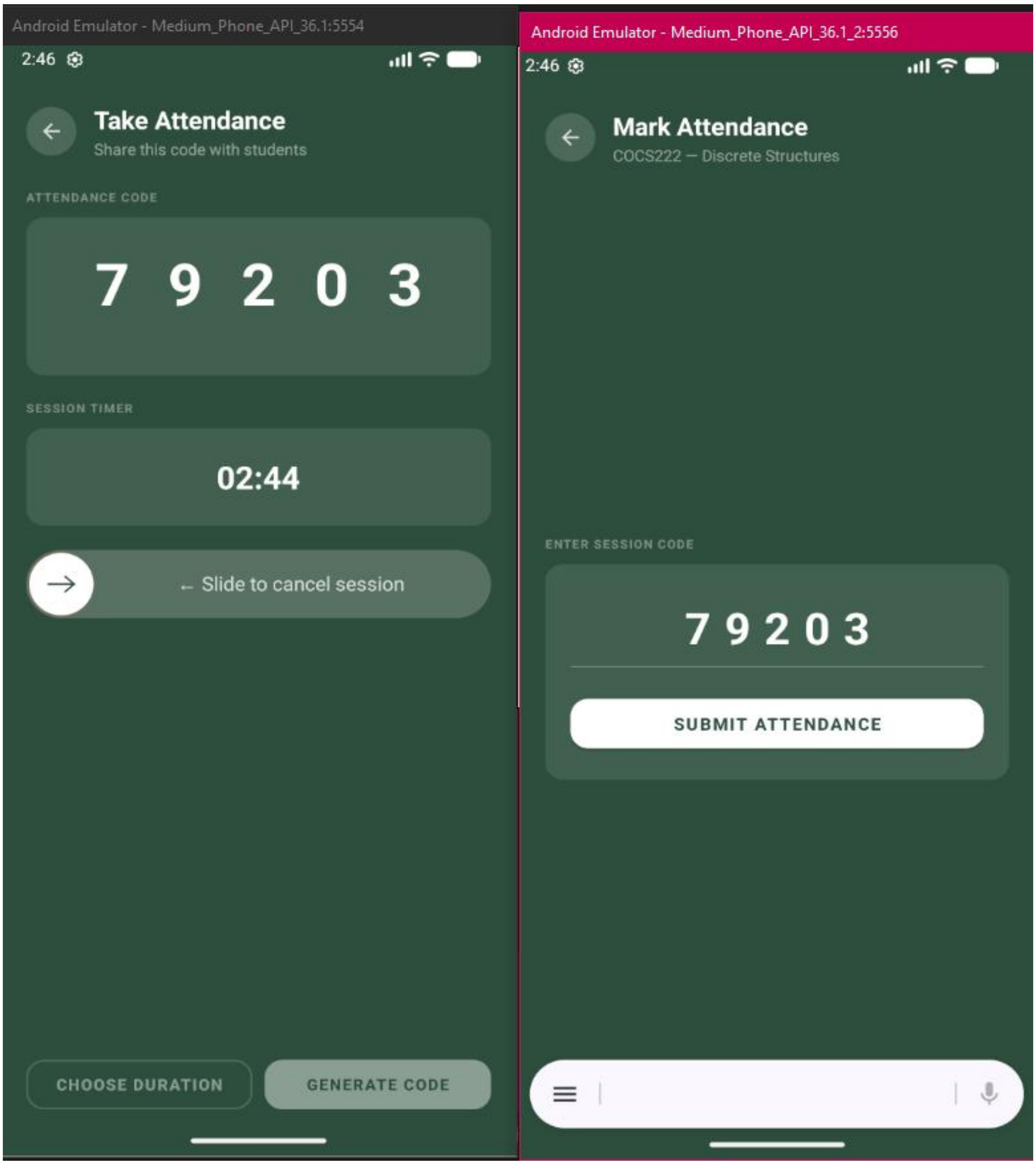

Figure 6.3 Attendance Session

6.3.4 Attendance History

The attendance history screen provides a detailed record of attendance for both instructors and students.

The instructor's session history displays a summary of total, present, late, and absent counts, along with a list of all enrolled students and their attendance status for that session.

The student's attendance history displays their overall present, late, and absent counts across all sessions for a specific course, with each session entry showing the date and attendance status.

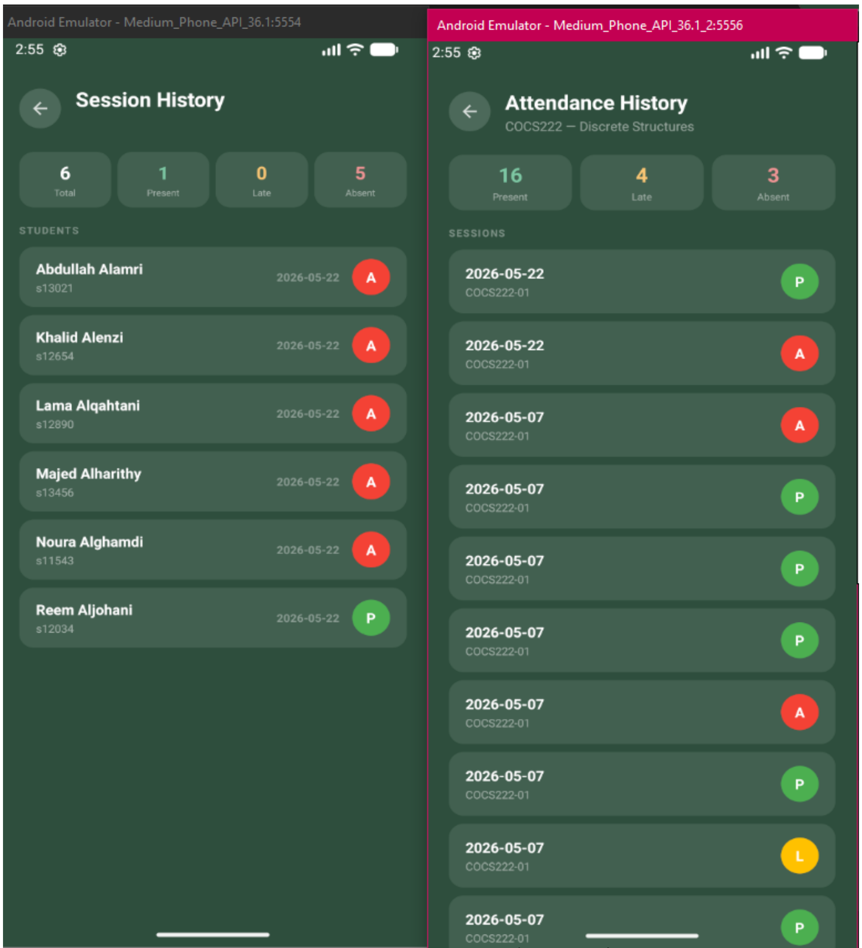

Figure 6.4 Attendance History

6.4 Conclusion

The testing phase of the mobile-based attendance system successfully verified that all components of the system function correctly both individually and as an integrated whole. Through the various testing stages including unit, integration, system, and user acceptance testing, all identified issues were resolved, resulting in a stable and reliable system.

The visual testing demonstrated through the application screenshots confirms that the system provides an intuitive and seamless experience for both students and instructors, from login through attendance management and history viewing.

The successful completion of the testing phase ensures that the mobile-based attendance system is ready for deployment and capable of meeting the attendance management needs of King Abdulaziz University.

CHAPTER 7

Discussion and conclusion

7.1 Introduction

This chapter discuss about achievement and final product that produced as well as the weakness and strength of the system for this system. There were discussions on improvement of system in future use.

7.2 Achievement and Product

We managed to make the process of scheduling the courses and get the desires of students specialization. This system is for FCITR members designed to make easier usability for the educational process.

7.3 Weakness of System

- The access control of users is managed manually by the admin of the system.

7.4 Suggestion Improvement in Future

- Expanding the system and giving the Instructor take full advantage of the system.
- The validity and flexibility of the system to be used in a regular semester rather than confine it just in the summer semester.
- Add notifications of news about the faculty (course registration, Specialization, Events etc.…)

7.5 Conclusion

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[1] N.A., Mashat, A.S. and AlBidewi, I., 2012. New chaos-based image encryption scheme for RGB components of color image. *Computer Science and Engineering*, *2*(5), pp.77-85.